Jasper Buntinx

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EDUCATION

Texas A&M University **Aug 2023 - 2027**
Bachelor of Science, Mechanical Engineering *College Station, TX*
• **GPA:** 3.31 / 4.0

EXPERIENCE

Interphase Transport Phenomena Lab (TAMU Nuclear Engineering) **Oct 2026 - Present**
Undergraduate Researcher *College Station, TX*
• Redesigning a parabolic flight payload containing 8-10 test modules to measure different applications of vortex gas-liquid separation in reduced gravity.
• Manufacturing fixtures and test hardware in-house using CNC machining, FDM printing, and MIG welding.
• Validating new carbon capture method, stripping CO2 from air through amine-based separation stages in series, integrated with a Rankine cycle.

Rochester Sensors (Amphenol Subsidiary) **May 2026 - Aug 2026**
Mechanical Engineering Intern *Dallas, TX*
• Programmed CNC machining workflows in CamWorks for 15+ production parts and dedicated multi-part fixtures, replacing inherited pattern-projected toolpaths to cut cycle times an average of 30% (182 → 34 min on highest-volume part) while holding all GD&T requirements.
• Resolved tolerance stack-up and assembly fit defects on in-development products in SolidWorks, validating designs by machining prototypes to released dimensions and correcting a source-design misalignment before mass production.
• Ran accelerated thermal cycling (−40 to +125 °C), salt ingress, and vibration qualification on pre-launch sensor assemblies, demonstrating 50+ year equivalent field life against a 30-year requirement.
• Programmed Universal Robots arm for vision-guided bin picking, using Keyence 3D scanning to identify randomly oriented parts and grip them from bulk bins before sorting them at 15–30 parts/min, automating the first station of a phased plan to redeploy operator hours

Waterloo Greenway Conservancy (Austin Creek Show) **May 2022 - Jan 2023**
Mechanical Engineering Intern *Austin, TX*
• Derived four unique SolidWorks assemblies from Odonata LLC artist samples, enabling production of large metal dragonfly structures with 14-ft+ wingspans.
• Led fabrication of prototyped and full-scale dragonfly structures by laser-cutting eight unique wing designs via Fusion 360 CAM, metal frame fabrication, welding 200+ joints, soldering wiring for four LED displays, and producing water-resistant electronics packaging.
• Cut peak wind/flood loads 30% by replacing hard plastic shells with a microporous ePTFE membrane, and reinforced against bending via wing reorientation, a wider steel base, and 300 lb ballast, verified in SolidWorks Simulation.

Petrobras (Geoenergia Lab) **Jun 2025 - Aug 2025**
Mechanical Engineering Intern *Balneário Camboriú, Brazil*
Optimized CUDA-based Lattice Boltzmann (LBM) CFD solvers using AI-assisted kernel profiling in NVIDIA Nsight Compute, resolving memory bottlenecks to push throughput past published open-source LBM benchmarks:
• Achieved 400%+ throughput gain (MLUPS) via FP64→FP32 migration and memory-access restructuring, validated against solver tolerances.
• Authored a Python mesh-generation script for fine-to-coarse grid interfaces, raising the stable peak Reynolds number 33% with no throughput loss.
• Diagnosed uncoalesced global memory access via Nsight Compute workload reports, remapping thread-to-data access for 60% better warp efficiency.
• Cut peak shared memory 35% (56KB→36KB) after identifying redundant data staging and enabling aggressive reuse across thread blocks.

PROJECTS

Autonomous Laundry-Picking Cable Claw (Prototype) **May 2026 - Present**
• Designed a ceiling cable robot that picks up laundry, using four NEMA 17 winches to steer a camera claw by inverse kinematics and YOLO-World vision to locate clothes; modeled slide-in winch stations and a servo-geared four-bar claw in SolidWorks.
• Engineered electrical system: an ESP32/A4988 controller per winch on a fused 12 V bus, a Wi-Fi ESP32 claw on LiPo power, and a KiCad PCB routed for automated SMD assembly.
• Directed AI-assisted development (Claude Code) of distributed motion control, choosing one controller per winch with host-planned paths and timed segments, verified in simulation and current testing.
• Fabricated 10+ PETG parts and aluminum heat fins machined on a manual mill, lowering calculated motor temperature inside vented shells from 51 to 41 °C.

Automatic Retractable RFID Bike Lock (Prototype) **Dec 2025 - Present**
• Designed a frame-mounted, NFC-unlocked bike lock with a retractable steel cable: 15-part 3D-printed clamshell assembly integrating a solenoid latch, PN532 RFID reader, Arduino Nano, Li-ion power system, and a spring-loaded cable spool.
• Manufactured 3D-printed PETG prototype housing with TPU compliant liner fitting Ø27–46 mm frames; now iterating toward a CNC-milled aluminum version with DFM constraints designed in from the start.

PLA Filament Recycler | Aggies Create (Ecofil Team) **Aug 2025 - May 2026**
• Fabricated functional prototypes of a plastic shredder, dehydrator, and extruder through 3D printing (OrcaSlicer), CNC milling (SolidWorks CAM), laser cutting (Lightburn), soldering, and MIG welding, providing working models for performance testing and design iteration.
• Held extruded filament diameter to 1.75±0.10 mm using closed-loop control, a linear Hall-effect sensor driving a stepper puller that corrects draw rate mid-run.

SKILLS

Software: SolidWorks, CSWA Certified, AutoCAD, OnShape, SolidWorks Simulation, NX, ANSYS, MATLAB, ParaView, Nvidia Nsight Compute, MS Office Suite, SolidWorks CAM, OrcaSlicer, Fusion 360 CAM, Lightburn, Mastercam
Programming: CUDA, C++, Python, Java, HTML, Arduino/Embedded C
Fabrication: GD&T, FEA, 3D printing, Woodworking, Metalworking, Soldering, MIG Welding, Injection Molding, CNC, Hand tools

EXTRACURRICULAR

One Army Men's Organization **Aug 2025 - Present**
Special Events Logistics Chair
• Led large event construction on $30K+ annual budget, taking builds from design/mockup through full fabrication w/ help from 80+ members for final assembly.
• Work year-round on sponsorship acquisition, obstacle construction, and promotion for Gladiator Dash, a 501(c)(3) 5K mud run raising $180K+ annually.